

Decomposing habituais

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1 Introduction

Starke (2021) defends a syncretism approach for languages that do not formally distinguish imperfective forms expressing progressive and habitual functions. Under this approach, progressives are structurally more complex, and are derived from habituais. In this chapter, we extend this general approach and argue that habituais must be further decomposed, enriching the functional sequence. In particular, we show that some habituais are exclusively used to report incidental generalizations whereas others are reserved for rule-like generalizations. As predicted, some languages such as Turkish morphologically distinguish the two functions, while others such as English work with a syncretic form. An immediate consequence of our analysis is that the progressive can only be syncretic with the habitual form that is employed to report incidental generalizations, unless the two habitual types are syncretic.

Habitual forms are observed to be syncretic with progressives in a number of typologically different languages (Comrie 1976, cf. Dahl 1985). For example, in Turkish, Laz and Czech, progressivity and habituality can be expressed with the same phonological form as illustrated in (1)-(3).

(1) *Turkish*

- a. *Aslı şu anda koş-uıyor.*
Aslı this moment run-IMPF
'Aslı is running right now.' (Progressive)
- b. *Aslı her gün koş-uıyor.*
Aslı every day run-IMPF
'Aslı runs every day.' (Habitual)

(2) *Laz*

- a. *k'oçi-k k'ata ndğas oxori ts'opx-um-s.*
man-ERG every day house build-IMPF-PRS.3SG
'The man builds houses every day.'
- b. *k'oçi-k hus oxori ts'opx-um-s.*
man-ERG now house build-IMPF-PRS.3SG
'The man is building a house now.'

(3) *Czech* (Starke 2021)

- Sekám dřevo.*
I.chop wood
'I am chopping/chop wood.' (Progressive or habitual)

In addition to regular habituality, Starke (2021) identifies an intermediate reading that he calls the *medial*. The medial is a type of habitual that spans over a time interval that is relatively closer to the reference time. These are forms modified by adverbs such as *recently* or *lately*.

- (4) a. Aslı is not eating well recently.
b. Aslı is reading this book lately.

Semantically, the medial shows distinct behavior from run-of-the-mill habituals in a number of properties. For example, run-of-the-mill habituals cannot be modified by adverbs such as *recently* or *lately* in English whereas the medial can.

- (5) #Aslı reads this book lately/recently.

Second, as Starke (2021) observes, it can only refer to an eventuality that is not distant from the reference time. For example, in (6), the eventualities that span over a distant period of time are reported to be odd whereas the neutral form without a temporal modifier is understood to assert an eventuality that has been occurring lately as shown in (7).

- (6) a. Salvador is not eating well lately.
b. ? Salvador is not eating well, this past year.
c. ?? Salvador is not eating well, over the past 10 years. (Starke 2021)

- (7) Aslı is not eating well.

By contrast, run-of-the mill habitual forms typically describe life-long properties as in (8).

(8) Aslı eats well.

Morphosyntactically, Starke (2021) observes that the medial is syncretic with the progressive form in some languages whereas it is syncretic with the habitual in others. For example, while the medial is expressed with the progressive in English as shown in the examples above, in Formal Dutch, it is expressed with the habitual forms shown in (9). Notice that the medial reading is not available with the regular habituality in English.

(9) a. *Ik ben aan het eten.*

I am at the eat

‘I am eating.’

b. *Ik eet deze dag niet genoeg.*

I eat these day not enough

‘I am not eating enough lately.’

c. *Ik eet chocolade.*

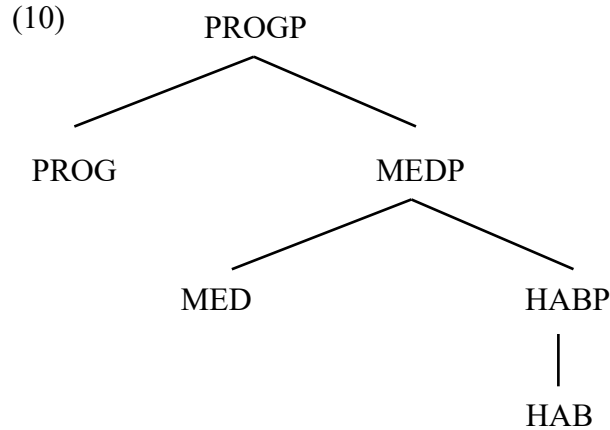
I eat chocolate

‘I eat chocolate (when I’m bored).’

(Starke 2021)

Under the approach that Nano-syntax assumes to account for syncretism (Caha 2009; 2013; Pantcheva 2010; Caha et al. 2022), the syncretism between the medial and habitual on the one hand, and the medial and progressive on the other suggests that the features that give rise to the progressive, medial and habitual are contiguous on the functional sequence (*fseq*). Based on this,

Starke (2021) proposes the *fseq* in (10) that builds progressive on top of medial and the medial on top of habitual.



Intuitively, the medial is formed by restricting the unrestricted time span expressed by the habitual to a recent time interval. Accordingly, the progressive is formed with the further restriction of the reference time. This analysis proposes to derive the meaning of the progressive from the medial, and the medial from the habitual. In this study, we aim to further decompose the habitual, and thus enrich the *fseq* proposed by Starke, based on data from a number of languages, mainly Turkish.

The outline of this chapter is as follows. In Section 2, we are going to present the semantic and morphosyntactic motivations to split **HAB** further. In Section 3, we flesh out the enriched *fseq* that we are proposing that reflects the further decomposition of the habitual. In Section 4, we discuss the crosslinguistic implications of our model. Section 5 concludes this chapter.

2 Characterizing habituality

In a number of typologically distinct languages, there are two morphemes marking habitual eventualities. For example in Turkish, in addition to the progressive morpheme as in (11a), the so-called “aorist” can express present habitual actions as shown in (11b).¹

- (11) a. *Aslı her gün erken kalk-ıyor.*
 Aslı every day early wake.up-IMPF
 ‘Aslı wakes up early every day.’
- b. *Aslı her gün erken kalk-ar.*
 Aslı every day early wake.up-AOR
 ‘Aslı wakes up early every day.’

The same two-way marking is available in Hinuq. According to Forker (2013), Hinuq, a Nakh-Daghestanian language, can express habituality with what she calls *habitual present* and *general tense*. The habitual present is marked with the suffix *λ’os* as in (12a) whereas the general tense is unmarked in Hinuq as shown in (12b).

- (12) a. *raḷad-li-š xan c’aq’ razi.iq-λ’os goḷ [λ’erba*
 sea-OBL-GEN1 khan(I) very be.pleased.I-HAB be guest
Ø-aq’e-yλ’o] hayloy eser-ho goḷa žo=n toλ-λ’os goḷ

¹ In Turkic linguistics, the aorist is a standard term for this morpheme although its meaning does not have to do with past eventualities as in other languages.

I-come-SIM he.ERG ask-ICVB be.PTCP thing=and give-HAB be

‘When a guest comes the sea king is very pleased and he gives him everything
that he asks for.’ (Forker 2013: 213)

b. [keč’ cax-el-mez] de Ø-iči-l-me
song write-POT-PURP.NEG I I-be-POT-NEG

‘I (masc.) am not able to write poems.’ (Forker 2013: 205)

Forker (2013) indicates that the habitual present expresses habitual or characteristic situations whereas the general tense expresses rule-like generalizations and characterizations as well as generic statements. Crucially, she also indicates that “the Habitual Present is generally not used in statements expressing rules” (Forker 2013:214).

Importantly, this general description to differentiate the two forms is also associated with different types of habituals mentioned in the literature. For one, it has been observed that forms used to make habitual generalizations that can also express rule-like generalizations and characterizations can have future reference (Krifka et al. 1995; Carlson 2006). In Turkish, among the two habitual markers, only aorist (-*Ar* habitual henceforth) can express habitualities necessarily projecting into the future. This is diagnosed by the fact that the future reference of the generalization expressed by -*Ar* habituals cannot be canceled as shown in (13b) whereas it can be if the generalization is expressed with the progressive, namely -*Iyor* (-*Iyor* habituals henceforth) as in (14b). The contrast between the two habitual forms shows that the generalizations reported by -*Iyor* habituals are *coincidental*, not rule-like.

(13) a. *Bu kasaba sakinleri şekersiz sakız çiğn-er.*

this town residents sugarless gum chew-AOR

‘Residents of this town chew sugarless gum.’

- b. *#ama gelecek kuşakla bu değişebilir.*

‘but this may change with the new generation.’

- (14) a. *Bu kasaba sakinleri şekersiz sakız çiğn-iyor.*

this town residents sugarless gum chew-IMPF

‘Residents of this town chew sugarless gum.’

- b. *ama gelecek kuşakla bu değişebilir.*

‘but this may change with the new generation.’

Another property that differentiates habitual types derives from whether the eventuality expressed by the habitual form has to have been actualized or not (Cable 2020; 2022). We observe that in Turkish, *-Ar* habituals, by virtue of them making rule-like generalizations, can express unactualized eventualities whereas there needs to be at least one occurrence of the relevant eventuality for *-Iyor* habituals to express true sentences. For example, in the context provided in (15a), the *-Ar* habitual in (15b) is able to state what would happen to a person consuming a particular plant, even though there has been no case of a person consuming that plant. However, the same context renders the *-Iyor* habitual infelicitous given the requirement that it express actualized generalizations.

- (15) a. CONTEXT: You have produced a new type of plant. Even though it has never been consumed or touched by anyone, you know that it is poisonous to people

because of its chemical composition. Hence, you say:

- b. *Bu bitki insanı zehirl-er.*
this plant person.ACC poison-AOR
'This plant poisons people.'

- b. *#Bu bitki insanı zehirl-iyor.*
this plant person.ACC poison-IMPF
'This plant poisons people.'

Needless to say, if there has been an occurrence of the plant poisoning people, then (15b) would also be felicitous. Such a context is provided in (16).

- (16) a. CONTEXT: You produced a new type of plant. Upon touching that plant,
a friend gets poisoned. Just as another friend of yours is about to touch it, you say:
- b. *Dokun-ma! Bu bitki insanı zehirl-iyor.*
touch-NEG this plant person.ACC poison-IMPF
'Do not touch it! This plant poisons people.'

Crucially, Turkish is not alone in differentiating between actualized and non-actualized eventualities with its habitual forms. Cable (2022) shows that the same distinction applies in Tlingit as well. In Tlingit, only what he calls "The Imperfective Mode" can express non-actualized habitualities whereas "The perfective habitual" is reserved for actualized ones as in (17).

- (17) a. Scenario (Based on Green 2000):

We just bought a new coffee machine. **It's never before been used.** But this is a great model of coffee machine. Everyone agrees that the coffee made by this model is great.

- b. *Yá yées aa washéen kúnáx linúktsi*
 DEM new PART machine very IMPRV.3SG.sweet.REL
coffee awe al.úkx
 coffee FOC 3O.IMPRV.3sg.boil.REP

‘This new machine boils very sweet coffee.’ (SE)

- c. *(#) Yá yées aa washéen kúnáx linúktsi*
 DEM new PART machine very IMPRV.3SG.sweet.REL
coffee awe ool.úkch
 coffee FOC 3O.HAB.PRV.SG.boil

Speaker comment: “No. That means that you’ve used it.” (SE) (Cable 2022:14)

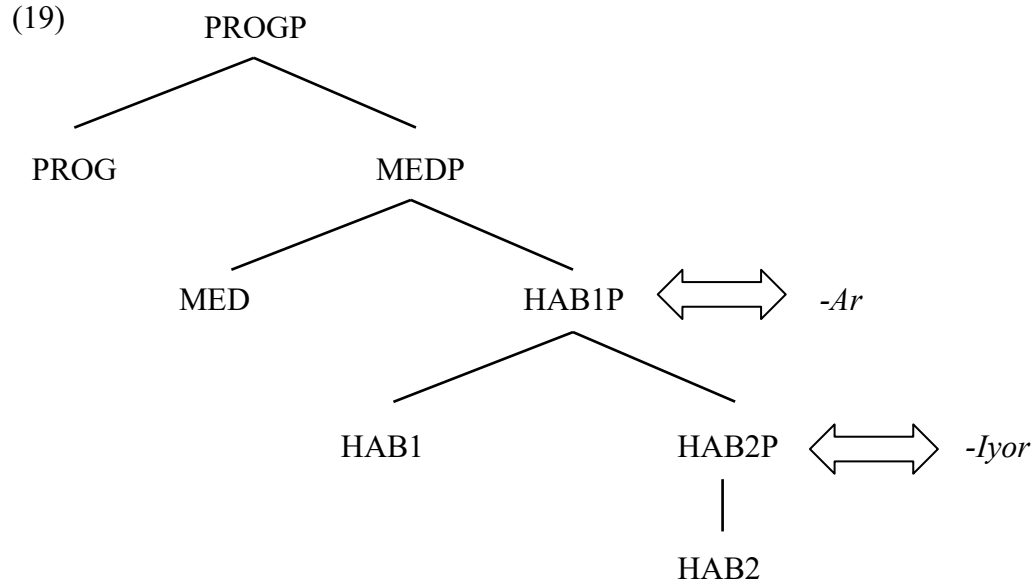
‘This machine crushes oranges.’ (Filip 2018 as cited in Cable 2022)

The fact that such semantic differences correspond to distinct lexicalizations in a number of typologically different languages suggests that habituality is not a single feature in the functional sequence. It appears that the two semantic contrasts that we have seen so far cluster together. In other words, if a habitual form in a given language can only express actualized events, then that form will fail to make rule-like generalizations that project into the future (hence they can only express incidental generalizations). However, if a habitual form in a given language can express non-actualized events, then that form should be able to make rule-like generalizations that project into the future. Based on this clustering of properties concerning distinct habitual forms, we argue

that there are at least two features associated with habituality in the functional sequence and further argue that one form builds on the other. In the following section, we model our empirical observations.

3 Enriching the functional sequence

Our empirical observations indicate that two distinct features may be involved in building habituals. At least one feature must give rise to the type of habitual meaning that allows reporting rule-like generalizations that necessarily project into the future and that can describe non-actualized generalizations. In Turkish, these are the semantic properties associated with what the aorist (*-Ar* habituals) lexicalizes. Let us assume that generating this meaning requires the feature HAB1. We have also shown that there is an additional type of habitual meaning that can be distinctly marked across languages and that expresses incidental habits. In Turkish, this corresponds to the type of habituality lexicalized by the progressive marker *-Iyor*. Let us assume that generating this meaning crucially requires the use of the feature HAB2. Under the assumption that one habitual meaning is structurally more complex and is built on top of the other, one can modify the functional sequence proposed in Starke (2021) in two ways. Either the rule-like habituals are derived on top of the incidental ones or vice versa. In the former case, we would expect the functional sequence in (19), where HAB1 is on top of HAB2.



Let us recall that in the model presented in (14), HAB1P corresponds to habituais characterized by rule-like expressions, and it is derived by merging HAB1 with HAB2P. Then, in this model, HAB1P would be lexicalized by *-Ar* whereas HAB2P alone would be lexicalized by *-Iyor* in Turkish. However, this prediction of the model is problematic because although both *-Ar* and *-Iyor* can express habitual events as shown in (20), there are crucial differences between their properties. *-Ar* in Turkish can neither express the medial reading nor the progressive reading while both of these readings are expressed by *-Iyor* in Turkish. These claims are substantiated by the data contrasts between (21) and (22).

- (20) a. *Aslı her gün erken kalk-ar.*
 Aslı every day early wake.up-AOR
 ‘Aslı wakes up early everyday.’
- b. *Aslı her gün erken kalk-ıyor.*
 Aslı every day early wake.up-IMPF

‘Aslı wakes up early everyday.’

- (21) a. #*Aslı son zamanlarda erken kalk-ar.*

Aslı late times early wake.up-AOR

Int(ended): ‘Aslı has been waking up early lately.’

- b. #*Aslı şu anda çok sakız çiğn-er.*

Aslı this moment much gum chew-AOR

Int: ‘Aslı is chewing a lot of gum now.’

- (22) a. *Aslı son zamanlarda erken kalk-ıyor.*

Aslı late times early wake.up-IMPF

‘Aslı has been waking up early lately.’

- b. *Aslı şu anda çok sakız çiğn-iyor.*

Aslı this moment much gum chew-AOR

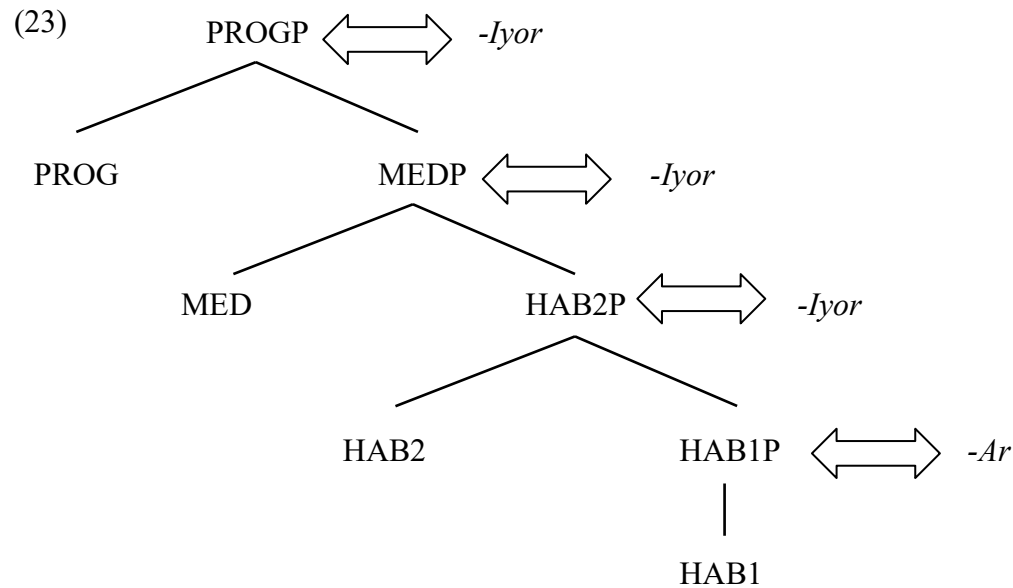
‘Aslı is chewing a lot of gum now.’

Given these observations, the functional sequence entertained in (19) would generate an *ABA pattern as illustrated in Table 1. Therefore, we rule out the possibility that generic habituals are derived from incidental ones.

	HAB2	HAB1	MED	PROG
Turkish	<i>-Iyor</i>			
	<i>-Ar</i>			
	<i>-Iyor</i>			
	<i>-Iyor</i>			

Table 1 An illustration of *ABA pattern

Hence, we argue that incidental habituals are derived from generic ones. Additionally, we propose that HAB2 is built on top of HAB1, which derives the correct lexicalization patterns for Turkish as illustrated in (23).



Accordingly, if a language has two distinct habitual forms, we predict that the form that can be used to report generalizations about unactualized eventualities and rule-like generalizations that

project into the future lexicalizes HAB1, while the habitual form that can do neither lexicalizes HAB1 *and* HAB2 (and higher features if it also does medial and progressive).

4 Attested syncretism patterns across languages

Given the functional sequence that we have proposed, our analysis predicts various types of syncretisms that will further demonstrate which features are contiguous in *fseq*. We will indicate the attested syncretism patterns in the form of a lexicalization table, as illustrated for Turkish below in Table 2.

	HAB1	HAB2	MED	PROG
Turkish	<i>-Ar</i>			
	<i>-Iyor</i>			
	<i>-Iyor</i>			
	<i>-Iyor</i>			

Table 2 Habitual to progressive patterns in Turkish

4.1 English

Given that the progressive form is reserved for expressing the medial and presently ongoing events in English, HAB1 and HAB2 must be syncretic, being lexicalized by the so-called *Simple Present Tense* in English as shown in the table in Table 3.²

² Note that the cells of lexicalization tables often refer to multi-morphemic forms or simply mention the name of a particular ‘paradigm’ such as Simple Present, which, in English for

	HAB1	HAB2	MED	PROG
English	<i>Simple Present</i>			
	<i>Simple Present</i>			
	<i>-ing</i>			
	<i>-ing</i>			

Table 3 Habitual to progressive patterns in English

Our analysis of habituals has obvious predictions for languages like English, where HAB1P and HAB2P are lexicalized by the same form. For one, we predict that in English, the Simple Present can not only do things that Turkish *-Ar* does, but should also be able to do things that Turkish *-Ar* cannot. This is because *-Ar* can only lexicalize up to HAB1 in the *fseq* whereas the Simple Present may lexicalize either HAB1 alone or HAB1 and HAB2 together.

In the discussion that follows, we show that this prediction is borne out. First, let us recall that the Turkish aorist expresses rule-like generalizations that describe the inherent properties of an entity. We exemplified this property with the example in (13), repeated in (24), where the villagers do characteristically chew sugarless gum, which we identified to be a core property inalienable from them as understood from the infelicity of (24b) as a continuation to (24a).

example, requires the suffix /z/ under the third person singular and appears to be lexicalized by the root otherwise. Needless to say, when we need to provide the lexicons that derive these forms in the individual languages under discussion, we would need to take into consideration adjacent features in the structures along with any relevant forms of allomorphy. This is orthogonal to our purposes in this paper, hence we leave it to future work.

- (24) a. *Bu kasaba sakinleri şekersiz sakız çiğn-er.*
 this town residents sugarless gum chew-AOR
 ‘Residents of this town chew sugarless gum.’
- b. *#ama gelecek kuşakla bu değişebilir.*
 ‘but this may change with the new generation.’

Let us also recall that the same continuation is felicitous with (25), where the habituality is expressed by *-Iyor*, which minimally lexicalizes [HAB1 [HAB2]].

- (25) *Bu kasaba sakinleri şekersiz sakız çiğn-iyor.*
 this town residents sugarless gum chew-IMPF
 ‘Residents of this town chew sugarless gum.’

Considering that the Simple Present in English can lexicalize either HAB1 alone or HAB1 and HAB2 together, we expect the English example in (26) to be ambiguous between (24) and (25). In other words, it should be able to express what is conveyed by (24), which is exclusively expressed by *-Ar* in Turkish, in addition to conveying the meaning expressed by (25), for which the Turkish *-Ar* cannot be used.

- (26) Residents of this town chew sugarless gum.

It is obvious that (26) has a generic sense, where chewing gum is understood to be the defining property of the residents of the relevant town. Indeed, it has been extensively shown in the literature that the Simple Present tense in English can convey rule-like generalizations. A standard example to show the generic sense of the Simple Present in English is the one in (27), which can express a generalization about either typhoons in general or the relevant part of the Pacific.

(27) Typhoons arise in this part of the Pacific. (Carlson 1989)

A more compelling question is whether it has the meaning associated with *-Iyor* habituais in Turkish as in (25). The sentence in (26) can felicitously be followed by the continuation in (28). We take this to indicate that Simple Present Tense in English can also convey what is expressed with *-Iyor* habituais in Turkish.

(28) ...but of course this may change with the new generation.

Secondly, considering that *-Ar* habituais can only express rule-like generalizations, they should be odd with predicates that strictly express temporary properties. In other words, they should be incompatible with predicates that cannot be construed as describing an individual-level property of an entity in the sense of Carlson (1977). Indeed, it has been argued in the literature that the Turkish aorist is used to describe a permanent property of the entity in question (Yavaş 1979). In contrast, such predicates would be compatible with *-Iyor* given that it conveys incidental generalizations.. In fact, a verb like *çalış* ‘work’ (in the sense of *being employed at*) describes a temporary, stage-level property of an individual. Therefore, they are odd with *-Ar* habituais that

describe a rule-like generalization as shown in (29a) because a temporary working state of Aslı is not generally understood to be her defining/generic property. In contrast, they are perfectly grammatical with *-Iyor* habituals as shown in (29b).

- (29) a. #Aslı bu şirkette çalış-ır.
 Aslı this company.LOC work-AOR
 ‘Aslı works for this company.’
- b. Aslı bu şirkette çalış-ıyor.
 Aslı this company.LOC work-IMPF
 ‘Aslı works for this company.’

Not surprisingly, these verbs are compatible with the Simple Present Tense to describe the habitual status of the entity in question as we predict for English. This is shown in (30).³

³ This is unsurprising given that the Simple Present Tense can express both stage level and individual level generalizations. Kratzer (1986) points out that standing on a chair is a stage level property of a person whereas being tall is permanent, hence individual-level. Regardless, she reports that the same form can express one’s temporary or permanent ability to touch the ceiling. For example, in (34a), John’s standing on the chair temporarily enables him to be able to touch the ceiling whereas in (34b), he has the permanent ability to do so given his extraordinarily long arms. Crucially, in either case, the form responsible for expressing the generalization about the ability is the same. (English lacks the morphological distinction between habitual and progressive in combination with the ability modal *can* (Bhatt 2006)).

(i) (Stump 1985:41-43)

(30) Aslı works for this company.

In addition, there are verbs that can describe generic (rule-like) properties in some contexts but express incidental generalizations in others. For example, Aslı residing in Istanbul cannot be a generic generalization. Therefore, it is incompatible with *-Ar* habituals in Turkish. It can only be expressed with *-Iyor* as predicted by our analysis. The contrast between the two is shown in (31).

- (31) a. #*Aslı İstanbul'da yaş[a]-ar.*
Aslı İstanbul.LOC live-AOR
'Aslı lives in Istanbul.'
- b. *Aslı İstanbul'da yaş[a]-ıyor.*
Aslı İstanbul.LOC live-IMPF
'Aslı lives in Istanbul.'

Crucially, one could also make a generic statement with the verb *yaşa* 'live' concerning the habitat of a species. In this sense, the verb *yaşa* becomes compatible with *-Ar* habituals in Turkish as shown in (32).

-
- a. Standing on a chair, John **can** touch the ceiling.
- b. Having unusually long arms, John **can** touch the ceiling.

(32) *Bu flamingolar nehir kenarlarında yaş[a]-ar.*

this flamingos river side.LOC live-AOR

‘These flamingos live by the river.’

Importantly, the first sense of *yaşa* in (31) is compatible with an incidental generalization, and therefore can only be expressed by the realization of HAB1 and HAB2 together, as expected. However, the second sense of this verb reports a defining/characteristic property of the flamingo kind, which is expressible with the feature HAB1 only, lexicalized by *-Ar* in Turkish. Given that the Simple Present Tense in English can lexicalize HAB1 only, or HAB1 and HAB2 together, it can express an incidental generalization about Aslı’s living place as in (31b) as well as a generic statement about flamingos as in (32). This prediction of our model is also borne out as shown in (33).

(33) a. Aslı lives in Istanbul.

b. These flamingos live by the river.

Finally, there is one important formal discrepancy between how Turkish and English mark stative verbs. English stative verbs cannot tolerate progressive marking (Lakoff 1966; Dowty 1979). For example, a verb such as *like* describes a generalization about the psychological state of an individual, and it cannot be used with the progressive form while preserving this meaning. The relevant contrast is indicated in (34).

(34) a. Aslı likes books.

b. #Aslı is liking books.

This pattern is expected because the progressive form in English does not lexicalize habituals at any level. It can only express medials or currently ongoing events. Given that liking describes a generalization about the mental state of an individual, it requires habitual features only so as to preserve the meaning relevant to (34a). In contrast, in Turkish not only *-Ar* but also *-Iyor* can felicitously be used to generalize over someone's psychological state. This is because *-Ar* lexicalizes HAB1, and *-Iyor* syncretically expresses not only medial and regular progressive, but also incidental habituality built up by the combination of HAB1 and HAB2. The relevant empirical points are exemplified in (35).

- (35) a. *Aslı* *kitapları* *sev-er.*
 Aslı books.ACC like-AOR
 ‘Aslı likes books.’
- b. *Aslı* *kitapları* *sev-iyor.*
 Aslı books.ACC like-IMPF
 ‘Aslı likes books.’

4.2 Ingush

One of the most immediate consequences of our model is that there must exist languages where each of the four features in the *fseq* is lexicalized by a distinct form. Although we have not been able to find a language where this prediction is completely met, we have been able to partially

confirm our prediction with data coming from Ingush. Ingush is a Northeast Caucasian language spoken by people living in the Republic of Ingushetia according to Nichols (1997). Nichols (1997) reports that the so-called “Present Tense” expresses rule-like generalizations. In her words, “only the simple present is appropriate for timeless generalizations like ‘Water boils at 100° C.’” (1997: 263). In contrast, the progressive form is able to convey generalizations that are incidentally true. In this respect, Ingush is similar to Turkish, which lexicalizes HAB1 alone or HAB1 and HAB2 together using distinct forms. In (36), we present four examples from Ingush, where the Present Tense expresses rule-like generalizations. Let us note that Simple Present in Ingush is generally unmarked, which would suggest that the root forms are able to lexicalize HAB1, which is at least partially the case in English as well.

(36) Ingush (Nichols 1997: 246-247, 262)

- a. *Aaz haara denna pwieghazh jul.*
 1s.ERG every day.DAT dish.PL J.wash
 ‘I wash the dishes every day.’
- b. *Yz sixa lel.*
 3SG fast walk.PRS
 ‘S/he walks fast.’
- c. *Wai aara wa-otta-dicha xii shal-lu.*
 winter.ADV outsideDX-put-D.CS.CVTEMP water freeze-VZ.PRS
 ‘In winter, water **freezes** if it’s left outside.’
- d. *Cuo xii mol.*
 3S.ERG water drink.PRS

‘He drinks water.’

Hence, The Present Tense in Ingush is used to express rule-like characterizations. This is most obvious in example (36c) where a scientific fact is conveyed about water. Quite interestingly, Turkish also uses the aorist form, which corresponds to HAB1 in our model, to express the same generalization rather than *-Iyor*. This is illustrated in (37).⁴

- (37) a. *Kışın su dışarıda bırak-ıl-ır-sa don-ar.*
in.winter water outside leave-PASS-AOR-COND freeze-AOR
‘In winter, water **freezes** if it’s left outside.’
- b. #*Kışın su dışarıda bırak-ıl-ır-sa don-uyor.*
in.winter water outside leave-PASS-AOR-COND freeze-IMPF
Int: ‘In winter, water **freezes** if it’s left outside.’

Hence, the description in Nichols (1977) suggests that HAB1 only corresponds to the Present Tense in Ingush. In contrast, the medial is expressed by one of the progressive forms in the language as shown in (38).

⁴ The progressive form in (37b) can felicitously be uttered by a primary school student upon observing it for the first time.

(38) *Yz xii molazh vy.*

3S water drink.CVSIM D.PROG.PRS

‘He is drinking water (these days).’

(Nominative progressive with ‘be’.)

(Nichols 1977: 262)

Accordingly, we would expect three possibilities for the lexicalization of HAB2 within the nano-syntactic understanding of contiguity of features. It could be syncretic with HAB1, or it could be syncretic with MED. Nichols (1977) observes that the so-called “progressive” form is also used to indicate repeated events or tendencies. These are not necessarily rule-like generalizations. For example, in (39), the progressive describes an incidental property of the horse that has to do with its startling, not necessarily its generic characterization. Therefore, we believe that MED is syncretic with HAB2 in Ingush. However, let us note that this is an inference based on the description of Nichols (1977), and that we lack conclusive evidence for this claim.

(39) *Yz gour q'exkazh jy.*

DEM horse startle:PLC.CVSIM J.PROG

‘This horse startles (often).’

(Nichols 1977: 264)

Interestingly, the progressive form expressing events ongoing at the speech time has a different auxiliary form as shown in (40). The data suggest that Ingush lexicalizes PROG with a different entry than MED. In other words, they are not syncretic.

(40) *Yz xii molazh laatt.*

(Nominative progressive with 'stand') (Nichols 1977:)

Laz (South Caucasian) is a language that has a single imperfective category. As expected, its lexical entry can reach up to the PROG feature and will be used in all relevant contexts, as illustrated in (41).

- (41) *Laz* (Muhammet Bal, pers. comm.)
- a. *K'atu-pe-k mjalva ş-um-an.*
 cat-PL-ERG milk drink-IMP-3PL
 ‘Cats drink milk.’ (rule-like generalization)
- b. *Şana-k k'ata ndğas mjalva ş-um-s.*
 Şana-ERG every day milk drink-IMP-3SG
 ‘Şana drinks milk every day.’ (incidental generalization)
- c. *Şana-k ham ndğalepes opşa mjalva ş-um-s.*
 Şana-ERG these days much milk drink-IMP-3SG
 ‘Şana is drinking a lot of milk these days..’ (medial)
- d. *Şana-k hus mjalva ş-um-s.*
 Şana-ERG now milk drink-IMP-3SG
 ‘Şana is drinking milk right now..’ (progressive)

Given these observations, we propose the lexicalization table in Table 5 for Laz.

	HAB1	HAB2	MED	PROG
Laz	-um			
	-um			
	-um			
	-um			

Table 5 Habitual to progressive patterns in Laz

5 Summary and Future Directions

5.1 Summary

In this study, we have shown that habituality can be expressed in at least two ways. HAB1 expresses generic generalizations whereas the combination of HAB1 and HAB2 conveys incidental ones. We have presented evidence that the splitting of the habitals into two distinct features is well motivated not only by certain semantic distinctions but also by morphosyntactic contrasts observed in a number of languages. In particular, we have shown that some languages lexicalize HAB1 differently from the combination of HAB1 and HAB2. Among these are Turkish, Ingush, and Hinuq. We present a summary of the lexicalization patterns that we have observed throughout the paper in Table 6.

	HAB1	HAB2	MED	PROG
English	“Simple Present”			
	-ing			
Turkish	-Ar			
	-Iyor			
Ingush	“Present Tense”			
	“Verb”-azh dy/vy			

	<i>“Verb”-azh laat</i>
Laz	<i>-um</i>

Table 6 A summary of syncretic patterns across languages

The newly proposed functional sequence has a number of cross-linguistic consequences. For one, we could account for the progressive marking found on stative verbs in Turkish as opposed to the English forms. Second, we could explain why the Simple Present Tense is compatible with incidental generalizations as well as rule-like claims in English unlike what we find in languages such as Ingush or Hinuq. Additionally, the proposed hierarchy of features predicts there to be a language where each feature is distinctly lexicalized. We have partially confirmed this prediction with Ingush where HAB1P is distinct from HAB2P and MEDP while HAB2P and MEDP are distinct from PROGP. An immediate consequence of the splitting of HAB into two features is that the meaning of the hierarchically higher type of habitual (i.e., HAB2) is built on top of the lower one (HAB1).

5.2 Future Directions

5.2.1 HAB1 and Conditionals

An obvious property of English factual hypothetical sentences that have a present and future reference is that the antecedent of the conditional has the Simple Present Tense. This is illustrated in (42).

- (42) a. If you freeze the water, it turns into ice.
b. If you leave now, you will lose all you have earned.

Considering that the Present Tense in English can lexicalize solely HAB1 or HAB1 and HAB2 together, it is not obvious whether the antecedent of a conditional in cases like (42) only involves HAB1, HAB1 and HAB2 together or both. Since the use of the Present Tense in the antecedent is not always linked to a scientific fact as observed in (42b), it is hard to see just on the basis of English, which features are employed in the antecedents of factual conditionals .

We believe that our model splitting habituality into two can help us answer this question, given that we can actually find distinct forms for the relevant features in different languages. For example, we observe that the antecedent of a conditional is overtly marked with the aorist/-Ar habitual in Turkish as shown in (43). Importantly, the progressive marking is odd to describe the given situations as illustrated in (44).

- (43) a. *Su-yu dondur-ur-sa-n, buz ol-ur.*
 water-ACC freeze-AOR-COND-2SG ice become-AOR
 ‘If you freeze the water, it turns into ice.’
- b. *Şimdi gid-er-se-n, tüm kazan-dığ-ın-ı kaybed-er-sin.*
 now go-AOR-COND-2SG all earn-NMZ-POSS-ACC lose-AOR-2SG
 ‘If you leave now, you will lose all you have earned.’

- (44) a. *#Su-yu dondur-uyor-sa-n, buz ol-ur.*
 water-ACC freeze-IMPF- COND-2SG ice become-AOR
 Int: ‘If you freeze the water, it turns into ice.’
- b. *# Şimdi gid-iyor-sa-n, tüm kazan-dığ-ın-ı kaybed-er-sin.*

now go-IMPF- COND-2SG all earn- NMZ-POSS-ACC lose-AOR-2SG

Int: 'If you leave now, you will lose all you have earned.'

The oddity of the examples in (44) likely results from the fact that the only interpretation available when the antecedent of a conditional involves the progressive marking is the regular use of progressive describing an ongoing event at the time of utterance. Hence, the antecedent in (44b) is on a par with the progressive usage in the antecedent of conditionals in English, (45).

(45) If you are leaving now, (do not forget to leave the keys).

The fact that what is used in the antecedent of a conditional in Turkish is the form that corresponds to HAB1 suggests that English also uses HAB1 in those contexts. We hope that this type of reasoning allowed by the analysis at hand could shed some light on the structural/semantic makeup of factual conditionals.

5.2.2. HAB1 and the Future

An important issue that we leave for future research is whether HAB1 can be part of another functional sequence. In particular, there is a possibility that it is the lowest feature of another *fseq* in certain cases. An important property of HAB1P in all the languages where it is lexicalized distinctly from HAB2P is that the relevant form lexicalizing the former can refer to future eventualities, by expressing predictions, speaker intentions and so forth. For example, in Hinuq, the General Tense can make a future reference, that is akin to a prediction, as illustrated in (46).

(46) *Hinuq* (Forker 2013: 206)

- a. *hezzo haylu-s g^wan b-aq'e*
 then she.OBL-GEN1 rheumatism(III) III-come
 ‘(After climbing too many mountains), she will get rheumatism.’
- b. “*zek de haze-z y-ike-me*”=*λen eλi-yo*
 tomorrow I they.OBL-DAT II-see-NEG=QUOT say-PRS
 ‘They will not see me tomorrow.’

Interestingly, the forms lexicalizing HAB1P in Ingush and Turkish respectively have similar uses. For example, in (47a), Turkish makes an explicit reference to a future probability whereas in (47b), the speaker's intention regarding a prospective event is conveyed.

- (47) a. (*Herhalde*) *yarın yağmur yağ-ar.*
 probably tomorrow rain rain-AOR
 ‘It will (probably) rain tomorrow.’
- b. *Yarın tekne turuna gel-ir-im.*
 tomorrow boat tour.DAT come-AOR-1.SG
 ‘I will come to the boat tour tomorrow.’

Similarly in Ingush, the Present Tense can also refer to future eventualities. The relevant examples are illustrated in (48).

(48) Nichols (1997:247)

a. *T'aaqqa, Xaza jaaxar my xannavii Eaga-qaala xannuu yz,*

so call.PPL.NZ EMPH be.NW.V.=Q (place) be.NW.V 3S

dolccha bessa dwa-duuc aaz hwuona

D.be.PPL.OBL:FOC nature.OBL DX-D.tell.PRS 1S.ERG 2S.DAT

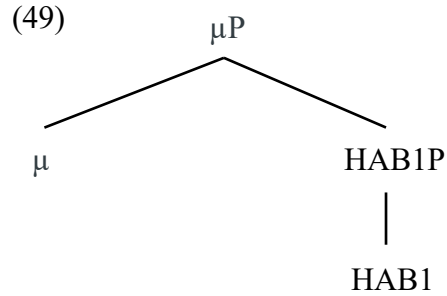
‘So there was this Xaza, he lived in Eaga-qaala, I’ll tell you all about it. (*dolccha*
bessa ‘in the appropriate way’)’ (0392B.1)

b. *Chei mol=ii wa?*

tea drink.PRS=Q 2S.ERG

‘Will you have some tea? (The speaker is pouring herself some tea and politely
offers tea to an office mate.)’

These examples are difficult to explain if only HAB1 is part of their structure, particularly because they refer to specific eventualities in the future, some of which make explicit use of future adverbials like *tomorrow*. Haspelmath (1998) observes the relationship between forms which correspond to HAB1 in our model and future reference. He points out that there is a similar diachronic process of developing future reference for such forms in a number of typologically distinct languages. Hence, it is likely that a ‘modal’ functional sequence builds on top of HAB1 in these languages as shown in (49). Obviously, the availability of this possibility predicts there to be languages where the addition of this new feature is marked differently from the lexicalization of HAB2P.



A second possibility would be that these are not readings distinct from the reading allowed by HAB1 alone but derive from the basic meaning of HAB1 in specific contexts. However, we could already discard this possibility since the simple Present Tense in English cannot express such future predictions as shown in (50). This implies that there are additional feature(s) building on HAB1 which allows reference to future eventualities.

(50) #Maybe, it rains tomorrow.

As we conclude this chapter, we leave for future work these intriguing questions and the implications arising from assuming that HAB1 can start another functional sequence.

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